Active vibration control of a cantilever beam using a piezo stack actuator

1. Introduction

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2. Modal analysis

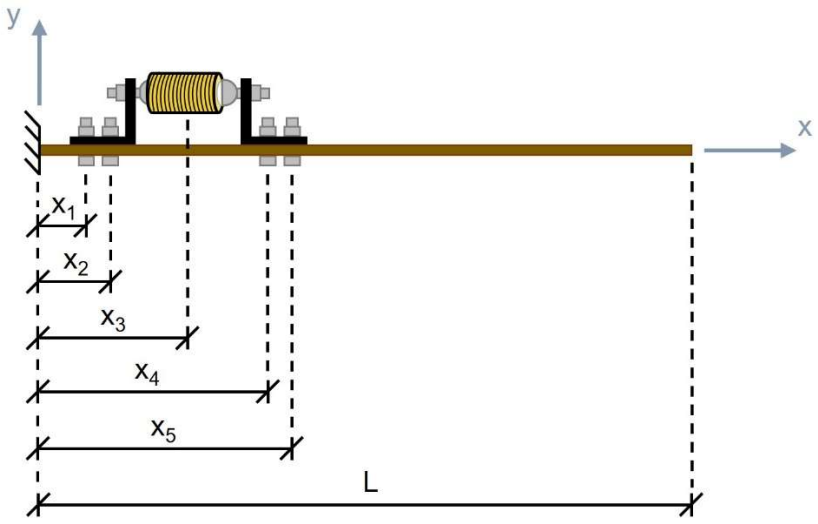


Figure 1: General configuration.

For a uniform beam, the Euler-Bernoulli equation in free vibration is:

$$\rho A \frac{\partial^2 y}{\partial t^2} + EI \frac{\partial^4 y}{\partial x^4} = 0. \tag{1}$$

Given that the system is linear time-invariant (LTI), we can assume a solution of the form:

$$y(x, t) = \psi(x)\, e^{j\omega t}. \tag{2}$$

This is because complex exponentials $e^{j\omega t}$ are eigenfunctions of any LTI system, i.e., they pass through the system unchanged in shape, only scaled and/or shifted.

Substituting (2) in (1):

$$\rho A\, \psi(x)\, j^2\omega^2\, e^{j\omega t} + EI \frac{\partial^4 \psi}{\partial x^4} e^{j\omega t} = 0,$$

$$-\frac{\rho A \omega^2}{EI}\psi(x) + \frac{\partial^4 \psi}{\partial x^4} = 0.$$

Defining $\beta^4 = \rho A\omega^2/EI$, the spatial equation for the mode shapes is:

$$\frac{\partial^4 \psi}{\partial x^4} - \beta^4 \psi = 0. \tag{3}$$

Assuming a solution of the form $\psi(x) = e^{rx}$, Eq. (3) becomes:

$$r^4 e^{rx} - \beta^4 e^{rx} = 0,$$

$$r^4 = \beta^4,$$

$$r = \sqrt[4]{\beta^4} = \sqrt{\sqrt{(\beta^2)^2}} = \pm\sqrt{\pm\beta^2},$$

$$r_1 = \beta,\ r_2 = -\beta,\ r_3 = j\beta,\ r_4 = -j\beta. \tag{4}$$

Since Eq. (1) is linear, the general solution is a linear combination of the four exponential solutions:

$$\psi(x) = \sum_{p=1}^{4} \kappa_p e^{r_p x} = \kappa_1\, e^{\beta x} + \kappa_2\, e^{-\beta x} + \kappa_3\, e^{j\beta x} + \kappa_4\, e^{-j\beta x}. \tag{5}$$

Eq. (5) has complex exponentials and exponentials with positive and negative arguments. Using Euler's formula and hyperbolic functions:

$$\begin{cases} e^{\theta} = \cosh\theta + \sinh\theta \\ e^{-\theta} = \cosh\theta - \sinh\theta \\ e^{j\theta} = \cos\theta + j\sin\theta \\ e^{-j\theta} = \cos\theta - j\sin\theta \end{cases},$$

the general solution can be expressed in real terms:

$$\kappa_1\, e^{\beta x} + \kappa_2\, e^{-\beta x} = \kappa_1(\cosh\beta x + \sinh\beta x) + \kappa_2(\cosh\beta x - \sinh\beta x) =$$

$$(\kappa_1 + \kappa_2)\cosh\beta x + (\kappa_1 - \kappa_2)\sinh\beta x\,,$$

$$\kappa_3\, e^{j\beta x} + \kappa_4\, e^{-j\beta x} = \kappa_3(\cos\beta x + j\sin\beta x) + \kappa_4(\cos\beta x - j\sin\beta x) =$$

$$(\kappa_3 + \kappa_4)\cos\beta x + j\,(\kappa_3 - \kappa_4)\sin\beta x\,.$$

Assuming κ_3 and κ_4 are complex conjugates, the constants $A = \kappa_1 + \kappa_2$, $B = \kappa_1 - \kappa_2$, $C = \kappa_3 + \kappa_4$, $D = \kappa_3 - \kappa_4$ can be defined and the imaginary constant is cancelled out. Equation (5) is therefore written as:

$$\psi(x) = A\cosh\beta x + B\sinh\beta x + C\cos\beta x + D\sin\beta x\,. \tag{6}$$

For a cantilever beam (fixed at x = 0 and free at x = L), the boundary conditions are:

$$\begin{cases} \psi(0) = 0 \\ \psi'(0) = 0 \\ \psi''(L) = 0 \\ \psi'''(L) = 0 \end{cases}. \tag{7}$$

Applying the first two boundary conditions from Eq. (7) to Eq. (6):

$$\psi(0) = A\cosh 0 + B\sinh 0 + C\cos 0 + D\sin 0 = 0,$$

$$C = -A, \tag{8}$$

$$\psi'(0) = \beta(A\sinh 0 + B\cosh 0 - C\sin 0 + D\cos 0) = 0,$$

$$D = -B. \tag{9}$$

Substituting Eq. (8) and (9) in (6):

$$\psi(x) = A(\cosh \beta x - \cos \beta x) + B(\sinh \beta x - \sin \beta x). \tag{10}$$

Applying the two last boundary conditions to Eq. (10), it becomes a linear system of equations in A and B:

$$\psi''(L) = \beta^2 [A(\cosh \beta L + \cos \beta L) + B(\sinh \beta L + \sin \beta L)] = 0,$$

$$\psi'''(L) = \beta^3 [A(\sinh \beta L - \sin \beta L) + B(\cosh \beta L + \cos \beta L)] = 0.$$

In matrix form:

$$\begin{bmatrix} \cosh \beta L + \cos \beta L & \sinh \beta L + \sin \beta L \\ \sinh \beta L - \sin \beta L & \cosh \beta L + \cos \beta L \end{bmatrix} \begin{bmatrix} A \\ B \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

For a homogenous linear system, a non-trivial solution exists if and only if the determinant of the matrix is zero. Thus:

$$(\cosh \beta L + \cos \beta L)^2 - (\sinh \beta L + \sin \beta L)(\sinh \beta L - \sin \beta L) = 0,$$

$$\cosh^2 \beta L + 2\cosh \beta L \cos \beta L + \cos^2 \beta L - \sinh^2 \beta L + \sin^2 \beta L = 0. \tag{11}$$

Applying the hyperbolic ($\cosh^2 \theta - \sinh^2 \theta = 1$) and trigonometric ($\cos^2 \theta + \sin^2 \theta = 1$) identities to Eq. (11), the characteristic equation of the system is obtained:

$$\cosh^2 \beta L - \sinh^2 \beta L^{\to 1} + \cos^2 \beta L + \sin^2 \beta L^{\to 1} + 2\cosh \beta L \cos \beta L = 0,$$

$$2^{\to 1} + 2^{\to 1} \cosh \beta L \cos \beta L = 0,$$

$$\cosh \beta L \cos \beta L + 1 = 0, \beta = \left(\frac{\rho A \omega^2}{EI}\right)^{1/4}. \tag{12}$$

Solving Eq. (12) numerically, the first roots are:

$$\begin{cases} \beta_1 L = 1.88 \\ \beta_2 L = 4.69 \\ \beta_3 L = 7.85 \\ \beta_4 L = 11.0 \\ \beta_5 L = 14.1 \end{cases}.$$

Considering the system parameters from Table 1, the natural frequencies $\omega'_k = \beta^2\sqrt{EI/\rho A}$ and $f'_k = \omega'_k/2\pi$ of the beam are shown in Table 2.

Table 1: Physical and geometric parameters of the beam.

L (m)	E (Pa)	I (m^4)	ρ (kg/m)	A (m^2)
0.55	3.5×10^9	6.3×10^{-10}	850	2.1×10^{-4}

Table 2: Natural frequencies of the beam.

k	ω'_k (rad/s)	f'_k (Hz)
1	40.9	6.50
2	256	40.7
3	717	114
4	1405	224
5	2322	369

It should be noted, though, that the formulation so far does not account for the contribution of the actuator's mass and stiffness to the response of the system. To incorporate it into the characteristic equation, the system can be modeled piecewise as shown in Table 3, where S_i and μ_i are the bending stiffness and the mass per unit length of each segment *i*. E_i, I_i, ρ_i, and A_i are the elastic moduli, moments of inertia, densities, and cross-section areas, with indices 1 and 2 referring to the beam and the actuator supports, respectively.

Table 3: Beam discretization.

i	Domain	S_i	μ_i
1	$0 \le x \le x_1$	E_1I_1	ρ_1A_1
2	$x_1 \le x \le x_2$	$E_1I_1 + E_2I_2$	$\rho_1A_1 + \rho_2A_2$
3	$x_2 \le x \le x_3$	E_1I_1	ρ_1A_1
4	$x_3 \le x \le x_4$	E_1I_1	ρ_1A_1
5	$x_4 \le x \le x_5$	$E_1I_1 + E_2I_2$	$\rho_1A_1 + \rho_2A_2$
6	$x_5 \le x \le L$	E_1I_1	ρ_1A_1

The piecewise Euler-Bernoulli equation is:

$$S_i \frac{d^4\psi_i}{dx^4} - \mu_i \omega^2 \psi_i = 0, \tag{13}$$

with the general solution being:

$$\psi_i(x) = A_i \cosh \beta_i x + B_i \sinh \beta_i x + C_i \cos \beta_i x + D_i \sin \beta_i x, \tag{14}$$

$$\beta_i = \left(\frac{\mu_i \omega^2}{S_i} \right)^{1/4}.$$

Since the beam vibrates as a single system, the frequency ω is common to all segments, thus becoming the main variable of the linear system.

The boundary and continuity conditions are shown in Table 4 and Table 5.

Table 4: Boundary conditions.

Clamped end	Free end
$\psi_1(0) = 0$	$\psi''_6(L) = 0$
$\psi'_1(0) = 0$	$\psi'''_6(L) = 0$

Table 5: Continuity conditions.

Deflection continuity	Slope continuity	Moment continuity	Shear continuity
$\psi_1(x_1) = \psi_2(x_1)$	$\psi'_1(x_1) = \psi'_2(x_1)$	$\psi''_1(x_1) = \psi''_2(x_1)\frac{S_2}{S_1}$	$\psi'''_1(x_1) = \psi'''_2(x_1)\frac{S_2}{S_1}$
$\psi_2(x_2) = \psi_3(x_2)$	$\psi'_2(x_2) = \psi'_3(x_2)$	$\psi''_2(x_2) = \psi''_3(x_2)\frac{S_3}{S_2}$	$\psi'''_2(x_2) = \psi'''_3(x_2)\frac{S_3}{S_2}$
$\psi_3(x_3) = \psi_4(x_3)$	$\psi'_3(x_3) = \psi'_4(x_3)$	$\psi''_3(x_3) = \psi''_4(x_3)\frac{S_4}{S_3}$	$S_4\psi'''_4(x_3) - S_3\psi'''_3(x_3) = m_a\omega^2\psi_3(x_3)$
$\psi_4(x_4) = \psi_5(x_4)$	$\psi'_4(x_4) = \psi'_5(x_4)$	$\psi''_4(x_4) = \psi''_5(x_4)\frac{S_5}{S_4}$	$\psi'''_4(x_4) = \psi'''_5(x_4)\frac{S_5}{S_4}$
$\psi_5(x_5) = \psi_6(x_5)$	$\psi'_5(x_5) = \psi'_6(x_5)$	$\psi''_5(x_5) = \psi''_6(x_5)\frac{S_6}{S_5}$	$\psi'''_5(x_5) = \psi'''_6(x_5)\frac{S_6}{S_5}$

Differentiating Eq. (14),

$$\begin{cases} \psi'_1(x) = \beta_i\left(A_i \sinh \beta_i x + B_i \cosh \beta_i x - C_i \sin \beta_i x + D_i \cos \beta_i x\right) \\ \psi''_1(x) = \beta_i^2\left(A_i \cosh \beta_i x + B_i \sinh \beta_i x - C_i \cos \beta_i x - D_i \sin \beta_i x\right) \\ \psi'''_1(x) = \beta_i^3\left(A_i \sinh \beta_i x + B_i \cosh \beta_i x + C_i \sin \beta_i x - D_i \cos \beta_i x\right) \end{cases},$$

the boundary and continuity conditions can be applied, leading to a system of equations of the form:

$$M(\omega) \cdot X = 0,$$

where M is a 24 x 24 coefficient matrix and X is the vector of unknowns given by:

$$M = \begin{bmatrix} M_1 & M_2 & M_3 \\ M_4 & M_5 & M_6 \\ M_7 & M_8 & M_9 \end{bmatrix},$$

$$M_1 = \begin{bmatrix} 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ ch_1(x_1) & sh_1(x_1) & c_1(x_1) & s_1(x_1) & -ch_2(x_1) & -sh_2(x_1) & -c_2(x_1) & -s_2(x_1) \\ sh_1(x_1) & ch_1(x_1) & -s_1(x_1) & c_1(x_1) & -sh_2(x_1) & -ch_2(x_1) & s_2(x_1) & -c_2(x_1) \\ ch_1(x_1) & sh_1(x_1) & -c_1(x_1) & -s_1(x_1) & -\frac{S_2}{S_1}ch_2(x_1) & -\frac{S_2}{S_1}sh_2(x_1) & \frac{S_2}{S_1}c_2(x_1) & \frac{S_2}{S_1}s_2(x_1) \\ sh_1(x_1) & ch_1(x_1) & s_1(x_1) & -c_1(x_1) & -\frac{S_2}{S_1}sh_2(x_1) & -\frac{S_2}{S_1}ch_2(x_1) & -\frac{S_2}{S_1}s_2(x_1) & \frac{S_2}{S_1}c_2(x_1) \\ 0 & 0 & 0 & 0 & ch_2(x_2) & sh_2(x_2) & c_2(x_2) & s_2(x_2) \\ 0 & 0 & 0 & 0 & sh_2(x_2) & ch_2(x_2) & -s_2(x_2) & c_2(x_2) \end{bmatrix},$$

$$M_2 = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -ch_3(x_2) & -sh_3(x_2) & -c_3(x_2) & -s_3(x_2) & 0 & 0 & 0 & 0 \\ -sh_3(x_2) & -ch_3(x_2) & s_3(x_2) & -c_3(x_2) & 0 & 0 & 0 & 0 \end{bmatrix},$$

$$M_3 = M_7 = \begin{bmatrix} 0 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 0 \end{bmatrix}_{8\times 8},$$

$$M_4 = \begin{bmatrix} 0 & 0 & 0 & 0 & ch_2(x_2) & sh_2(x_2) & -c_2(x_2) & -s_2(x_2) \\ 0 & 0 & 0 & 0 & sh_2(x_2) & ch_2(x_2) & s_2(x_2) & -c_2(x_2) \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix},$$

$$M_5 = \begin{bmatrix} -\frac{S_3}{S_2}ch_3(x_2) & -\frac{S_3}{S_2}sh_3(x_2) & \frac{S_3}{S_2}c_3(x_2) & \frac{S_3}{S_2}s_3(x_2) & 0 & 0 & 0 & 0 \\ -\frac{S_3}{S_2}sh_3(x_2) & -\frac{S_3}{S_2}ch_3(x_2) & -\frac{S_3}{S_2}s_3(x_2) & \frac{S_3}{S_2}c_3(x_2) & 0 & 0 & 0 & 0 \\ ch_3(x_3) & sh_3(x_3) & c_3(x_3) & s_3(x_3) & -ch_4(x_3) & -sh_4(x_3) & -c_4(x_3) & -s_4(x_3) \\ sh_3(x_3) & ch_3(x_3) & -s_3(x_3) & c_3(x_3) & -sh_4(x_3) & -ch_4(x_3) & s_4(x_3) & -c_4(x_3) \\ ch_3(x_3) & sh_3(x_3) & -c_3(x_3) & -s_3(x_3) & -\frac{S_4}{S_3}ch_4(x_3) & -\frac{S_4}{S_3}sh_4(x_3) & \frac{S_4}{S_3}c_4(x_3) & \frac{S_4}{S_3}s_4(x_3) \\ -S_3\beta_3^3 sh_3 x_3 & -S_3\beta_3^3 ch_3 x_3 & -S_3\beta_3^3 s_3 x_3 & S_3\beta_3^3 c_3 x_3 & S_4\beta_4^3 sh_4 x_3 & S_4\beta_4^3 ch_4 x_3 & S_4\beta_4^3 s_4 x_3 & -S_4\beta_4^3 c_4 x_3 \\ -m_a\omega^2 ch_3 x_3 & -m_a\omega^2 sh_3 x_3 & -m_a\omega^2 c_3 x_3 & -m_a\omega^2 s_3 x_3 & & & & \\ 0 & 0 & 0 & 0 & ch_4(x_4) & sh_4(x_4) & c_4(x_4) & s_4(x_4) \\ 0 & 0 & 0 & 0 & sh_4(x_4) & ch_4(x_4) & -s_4(x_4) & c_4(x_4) \end{bmatrix},$$

$$M_6 = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -ch_5(x_4) & -sh_5(x_4) & -c_5(x_4) & -s_5(x_4) & 0 & 0 & 0 & 0 \\ -sh_5(x_4) & -ch_5(x_4) & s_5(x_4) & -c_5(x_4) & 0 & 0 & 0 & 0 \end{bmatrix},$$

$$M_8 = \begin{bmatrix} 0 & 0 & 0 & 0 & ch_4(x_4) & sh_4(x_4) & -c_4(x_4) & -s_4(x_4) \\ 0 & 0 & 0 & 0 & sh_4(x_4) & ch_4(x_4) & s_4(x_4) & -c_4(x_4) \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix},$$

$$M_9 = \begin{bmatrix} -\frac{S_5}{S_4}ch_5(x_4) & -\frac{S_5}{S_4}sh_5(x_4) & \frac{S_5}{S_4}c_5(x_4) & \frac{S_5}{S_4}s_5(x_4) & 0 & 0 & 0 & 0 \\ -\frac{S_5}{S_4}sh_5(x_4) & -\frac{S_5}{S_4}ch_5(x_4) & -\frac{S_5}{S_4}s_5(x_4) & \frac{S_5}{S_4}c_5(x_4) & 0 & 0 & 0 & 0 \\ ch_5(x_5) & sh_5(x_5) & c_5(x_5) & s_5(x_5) & -ch_6(x_5) & -sh_6(x_5) & -c_6(x_5) & -s_6(x_5) \\ sh_5(x_5) & ch_5(x_5) & -s_5(x_5) & c_5(x_5) & -sh_6(x_5) & -ch_6(x_5) & s_6(x_5) & -c_6(x_5) \\ ch_5(x_5) & sh_5(x_5) & -c_5(x_5) & -s_5(x_5) & -\frac{S_6}{S_5}ch_6(x_5) & -\frac{S_6}{S_5}sh_6(x_5) & \frac{S_6}{S_5}c_6(x_5) & \frac{S_6}{S_5}s_6(x_5) \\ sh_5(x_5) & ch_5(x_5) & s_5(x_5) & -c_5(x_5) & -\frac{S_6}{S_5}sh_6(x_5) & -\frac{S_6}{S_5}ch_6(x_5) & -\frac{S_6}{S_5}s_6(x_5) & \frac{S_6}{S_5}c_6(x_5) \\ 0 & 0 & 0 & 0 & ch_6(L) & sh_6(L) & -c_6(L) & -s_6(L) \\ 0 & 0 & 0 & 0 & sh_6(L) & ch_6(L) & s_6(L) & -c_6(L) \end{bmatrix},$$

$$\begin{cases} ch_i(\theta) = \cosh \beta_i \theta \\ sh_i(\theta) = \sinh \beta_i \theta \\ c_i(\theta) = \cos \beta_i \theta \\ s_i(\theta) = \sin \beta_i \theta \end{cases},$$

$$X = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix},$$

$$X_1^T = [A_1 \quad B_1 \quad C_1 \quad D_1 \quad A_2 \quad B_2 \quad C_2 \quad D_2 \quad A_3 \quad B_3 \quad C_3 \quad D_3],$$

$$X_2^T = [A_4 \quad B_4 \quad C_4 \quad D_4 \quad A_5 \quad B_5 \quad C_5 \quad D_5 \quad A_6 \quad B_6 \quad C_6 \quad D_6].$$

The characteristic equation is:

$$\det[M(\omega)] = 0. \tag{15}$$

A code was developed in Python to solve Eq. (15) numerically and find the angular and linear natural frequencies, ω_k and $f_k = \omega_k/2\pi$, of the beam-actuator system. Table 6 shows the parameters used, where m_a is the actuator's mass. The positions x_1 to x_6 and L can be referred to in Figure 1.

Table 6: Physical and geometric parameters of the beam-actuator system.

E_2 (Pa)	I_2 (m^4)	ρ_2 (kg/m)	A_2 (m^2)	m_a (kg)	x_1 (m)	x_2 (m)	x_3 (m)	x_4 (m)	x_5 (m)	L (m)
2.0×10^{11}	1.0×10^{-11}	7800	3.0×10^{-5}	0.14	0.01	0.028	0.061	0.093	0.121	0.25

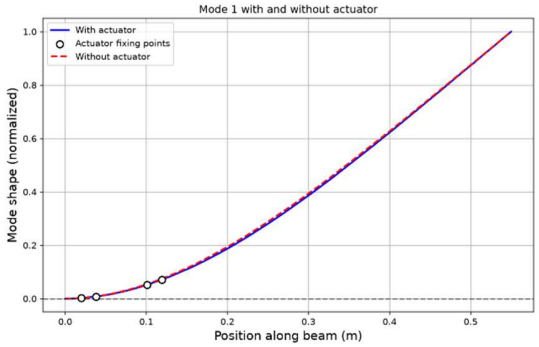
Table 7 presents the first five natural frequencies and compares them with the values found for the isolated beam (see Table 2).

Table 7: Natural frequencies of the beam-actuator system.

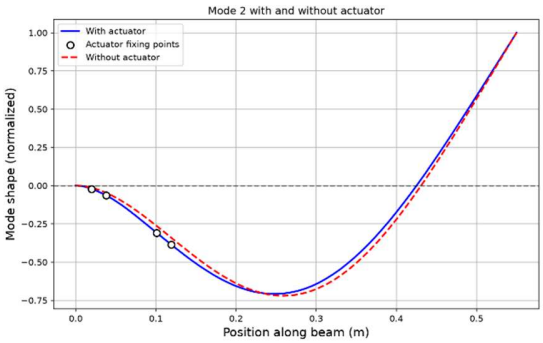
k	ω_k (rad/s)	f_k (Hz)	$(f_k - f'_k)/f'_k$ (%)
1	42.6	6.78	4
2	253	37.4	–8
3	487	77.5	–32
4	993	158	–29
5	1846	294	–20

The presence of the actuator introduced two competing mechanisms: additional bending stiffness, which raises natural frequencies, and additional mass, which lowers them. To comprehend which mechanism dominates in each vibration mode, leading to the variations in frequency shown in Table 7, the mode shapes must be examined.

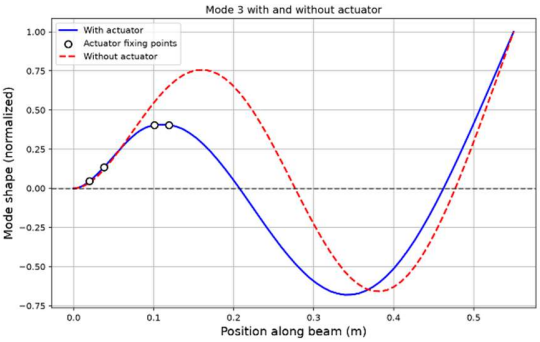
By applying the natural frequencies and system parameters from Table 6 and Table 7 to Eq. (14), the mode shapes $\psi_{k,i}(x)$ are obtained. Figure 2 shows the first five mode shapes of the system with and without the actuator normalized by the maximum modal amplitudes for visual comparison.



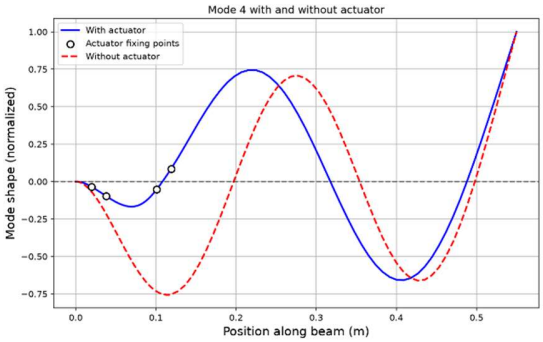
(a) Mode 1.



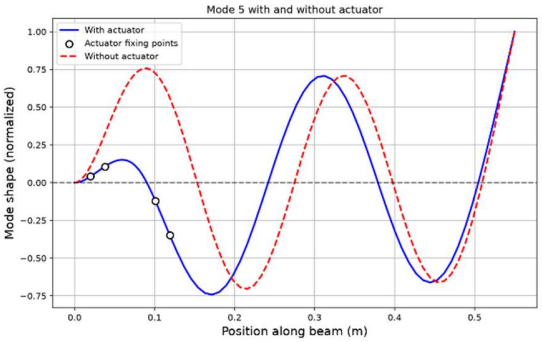
(b) Mode 2.



(c) Mode 3.



(d) Mode 4.



(e) Mode 5.

Figure 2: Comparison of mode shapes with and without the actuator.

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3. System identification

3.1. Mass-normalized mode shapes

The mode shapes $\psi_k(x)$ are eigenfunctions of Eq. (13) and therefore are defined only up to an arbitrary scaling factor, i.e., any scalar c_k multiplied $\psi_k(x)$ also satisfies the differential eigenvalue problem. To obtain a unique, physically meaningful set of mode shapes, a mass-weighted inner product of the mode shapes can be performed to define scalar constants c_k based on system parameters.

In the same way that the dot product of two vectors measures their overlap in Euclidean space, the inner product of two functions,

$$\langle f(x), g(x)\rangle = \int_a^b f(x)g(x)\,dx\,,$$

measures their overlap in a definite domain $a \le x \le b$ within the infinite-dimensional Hilbert space.

In the beam-actuator system, the mass-weighted inner product of a mode shape $\psi_k(x)$ with itself yields:

$$\langle \psi_k, \psi_k\rangle_\mu = \sum_{i=1}^{6} \int_{x_{i-1}}^{x_i} \mu_i \psi_{k,i}^2(x)\,dx + m_a \psi_{k,3}^2(x_3) = m_k, \tag{16}$$

where m_k is the modal mass of the vibration mode *k*. The modal mass represents the portion of the system's mass that contributes to its kinetic energy when vibrating in that mode. By weighting the mass distribution with the modal amplitudes and integrating in space, the modal mass converts the system into a simple, mathematically equivalent single-degree-of-freedom (SDOF) oscillator for that mode.

It should be noted, though, that the modal mass depends on the scaling of ψ_k. To avoid the dependency on arbitrary scales, the following normalization condition is imposed:

$$\langle \varphi_k, \varphi_k\rangle_\mu = 1, \tag{17}$$

where φ_k is a multiple of ψ_k of the form:

$$\varphi_k(x) = c_k \psi_k. \tag{18}$$

Substituting Eq. (18) into (17)

$$\langle c_k\psi_k, c_k\psi_k\rangle_\mu = 1,$$

Considering the linearity of Eq. (16), c_k^2 can be factored out:

$$c_k^2\langle \psi_k, \psi_k\rangle_\mu = 1,$$

$$c_k^2\, m_k = 1,$$

$$c_k = \frac{1}{\sqrt{m_k}}\,,$$

leading to the final form of the mass-normalized mode shape:

$$\varphi_k(x) = \frac{\psi_k(x)}{\sqrt{m_k}}\,. \tag{19}$$

3.2. Damping ratios

...

4. Control design

...

The system will have an accelerometer at the tip of the beam and a moment actuator near the fixed support. For such a single-input single-output (SISO) system, there are many...

4.1. Proportional-integral-derivative (PID)

...

4.2. Linear quadratic regulator (LQR)

...

4.3. Artificial neural networks (ANN)

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